
Peter (Bide) Huang
Mobile: +61 430 001 790
Company Website:
<https://www.hoversense.com.au/>

Email: hbd730@gmail.com

Summary

An enterprise software and solution architect with 15+ years of experience in software and system engineering, enterprise cloud applications, and real-time embedded systems.

Co-Founder and Director of a SaaS business, specializing in EDI integration for logistics and decision intelligence software. Proven ability to lead teams, build customer relationships, develop innovative solutions, and drive business growth through technical excellence and strategic leadership.

Highly driven to solve complex enterprise integration and industry-domain problems through modelling, solution design, and implementation for customers.

Experience

Endava Asia Pacific
Solution Architect / Integration Tech Lead

Melbourne, Australia
January 2025 – August 2026

Led Coal LSL Integration Platform solution architecture and product development in AWS. Defined integration standards, produced HLD/LLD specifications, and secured ARB approval. Shaped event-driven architecture and process-driven workflows, governed API design and management across technology streams, guided engineering teams through cross-team refinement and delivery of use cases, led a CQRS and Event-Sourcing PoC for Registry, and architected a solution for the Registry data migration tool.

- Led solution architecture and integration technology leadership across multiple technology streams, aligning API design, event-driven architecture, integration standards, and delivery practices.
- Guided engineering teams through cross-team refinement, architecture review, and delivery of integration use cases.

Technologies: AWS (Step Functions, Lambda, API Gateway, EventBridge, S3, DynamoDB, CloudWatch), GitHub Actions, .NET, Terraform, CQRS, Event Sourcing

Development Consultant

July 2024 – December 2024

Lead AI/ML initiatives and engage in technical presales and proof-of-concept development.

- Led a team of 5 to architect and deploy a cloud-based, real-time carry-on luggage detection and counting solution using computer vision and deep learning on AWS.
- Built an AI Financial Analyst tool leveraging Gemini Pro LLM with RAG architecture, streamlining document summarization and analysis workflows in capital markets.
- Achieved 3rd place in the Endava Australia Innovation Lab for an AI baggage-counting solution.

Technologies: AWS (EC2, S3, Lambda, SageMaker), Python, PyTorch, OpenCV, Google Gemini Pro, LangChain, RAG

Software/System Integration Lead

January 2023 – July 2024

Led software and system integration for Seeing Machine Smart Camera with Caterpillar in-vehicle control unit.

- Architected and implemented health and telemetry data collection, processing, and transmission from an in-vehicle Driver Safety System (DSS), integrating with Guardian Live, a cloud-based monitoring platform in AWS, via 4G network.
- Led development teams in Vietnam and Australia to deliver software releases by adopting Model-Based Systems Engineering (MBSE) to complement V-model systems engineering implementation.

Technologies: AWS (IoT Core, CloudWatch), C/C++, Python, 4G/LTE, MBSE, V-Model

Hover Sense Pty Ltd
Co-Founder and Director

Melbourne, Australia
January 2022 – Present

Built and commercialized a multi-tenant EDI integration solution platform from the ground up for national logistics companies, enabling seamless integration between ERP systems and trading partners' TMS/WMS platforms through AWS cloud technologies. Drove business growth through high-quality, cost-effective solutions and delivery.

- Developed an AI-powered fleet planning demo application to automate transport job allocation, generate optimized routes for a fleet of 150 vehicles, and assign jobs to drivers and trailers under real-world constraints involving 10,000+ variables. Improved truck utilization and demonstrated potential operational cost reductions of up to 15%.
- Collaborated with the Norwegian University of Science and Technology to develop quadcopter guidance and navigation simulation software for reinforcement learning research.

Technologies: AWS (Lambda, API Gateway, S3, DynamoDB, SQS, CloudFormation), Python, Node.js, React, EDI/X12, OR-Tools

SwarmFarm Robotics
Senior Software Engineer

Melbourne, Australia
Jun 2021 – January 2023

Reported to the CPO and held end-to-end responsibility for research and development of a robot route-planning software stack in C++. Led a team of three developers to develop and commercialize the cloud-based Farm Manager website.

- Researched and developed large-scale robot route-planning software, SwarmFarm Planning, for broadacre, orchard, and turf environments. Successfully integrated it with SwarmBots and the Farm Manager application and released it into production.
- Achieved a 98% path-planning success rate and reduced real-time robot traversal generation to 200 ms by defining KPIs and designing and deploying an evaluation server to run large-scale farm datasets in the cloud.
- Successfully scaled up the production workflow for traversal graph model generation in the cloud by designing a serverless application to generate field models in parallel using AWS API Gateway, Lambda, and S3.
- Significantly reduced manual effort, typically 20 person-hours for a 100-hectare field, in robot optimal work-plan creation by designing an algorithm to automatically generate path plans from surveyed geometries.

Technologies: C++, Python, AWS (API Gateway, Lambda, S3), React, PostgreSQL, Docker, QGIS

Bosch
Lead Software and System Engineer

Melbourne, Australia
Jun 2018 – Jun 2021

Reported to the Engineering Manager and held end-to-end responsibility for delivering software products and services for various customers, including the Victorian Government and OEMs. Defined engineering tasks according to project requirements, designed system software architectures, developed features, integrated algorithms, conducted testing, managed releases, organized customer meetings, and coordinated cross-regional collaborations.

- Designed and developed the prototype of electronic horizon for L2+ Advanced Driver Assist System, including horizon APIs for crowd-source map behaviour layer data extraction, ROS based horizon provider application, and RViz/DeckGL visualization tools.

- Successfully demonstrated a fully automated driving system for a \$2.5 million government-funded CAV trial project. Held end-to-end responsibility for mapping and localization, including GPS, camera, lidar, and radar sensor integration and calibration, data extraction and visualization, algorithm development, mapping activities, and road testing.
- Designed a novel approach for developing an HD planner map-creation pipeline and achieved centimetre-level localization accuracy through multi-beam lidar intensity calibration, multi-sensor calibration, and multi-modal localization performance optimization.
- Researched and developed a graph-based Visual SLAM algorithm with a stereo video camera for EKF sensor fusion, and created and deployed a multi-layer spatial data visualization website for analysis and critical decision-making.
- Successfully delivered an ultrasonic-sensor-based wade assist function for a JLR customer project, including embedded application design and implementation, sensor data filtering algorithms, simulation, and testing.
- Finished as the second runner-up in a machine learning business innovation competition by developing a prototype vision-based lane-marker detection system using semantic segmentation for road-quality analysis.

Technologies: C++, Python, ROS, RViz, DeckGL, OpenCV, PCL, Eigen, Ceres, Docker, lidar, radar, GPS, ultrasonic sensors

Blackmagic Design
Software Engineer

Melbourne, Australia
Jun 2014 – Jun 2018

Reporting to software engineering manager, responsible for real time embedded software development for high performance video post-production product, including drivers and application on micro-kernel based OS, API and multi-threaded cross platform video playback and capture applications and tools.

- Worked closely with other engineers to develop and release the world's smallest Ultra HD broadcast deck with professional 10-bit video/audio dual disk SD/UHS-II recording and playback.
- Served as lead engineer for the successful delivery of a Disney customer release for HyperDeck Mini Studio. Designed and implemented the Advanced Media Protocol, which enabled video playback synchronization, frame-accurate timeline control, and slow-motion control over serial and Ethernet.
- Served as lead engineer for the Video Assist 4K product and successfully delivered and demonstrated 3D LUT video color-correction preview and multilingual display and update features at CES 2017.

Technologies: C/C++, Objective-C, QNX, embedded RTOS, OpenGL, serial/Ethernet protocols, multi-threaded programming

Welling & Crossley
Electronics Engineer

Melbourne, Australia
Jan 2011 – Jun 2014

Reported to the Chief R&D Engineer and was responsible for researching and developing electronic schematics and PCBs, embedded software and device drivers for engine control units, Qt-based touchscreen controllers, variable-speed pump controllers, and remote controllers for irrigation and power-generation systems.

- Received an outstanding employee achievement award for the successful design and deployment of an engine-speed sensor for diesel and petrol power generators, which saved significant manufacturing costs.
- Successfully demonstrated an AS-2941-compliant touchscreen fire-pump controller to the company CEO and stakeholders.
- Performed critical troubleshooting on a half-million-dollar "Black Start" 500 kVA power generator for a Queensland water treatment plant.

Technologies: C, Qt, FreeRTOS, embedded systems, PCB design, Altium, LabView, PLC

Qualification

Certifications

- **Google:** Professional Cloud Architect Certificate *August 2024*
- **AWS:** Solution Architect Associate Certificate *February 2024*
- **Coursera:**
 - Google Project Management: Professional Certificate *February 2023*
 - Machine Learning Specialization *June 2020*
- **Udacity:** Self-Driving Car Nano Degree Certificate *Nov 2016 – Sep 2017*
- **Freiburg University:** Robot Mapping *September 2018*

Education

RMIT University <i>Graduate Diploma in Computer Engineering (High Distinction)</i>	Melbourne, Australia <i>September 2010 – November 2010</i>
--	---

RMIT University <i>Master of Electronics Engineering (High Distinction)</i>	Melbourne, Australia <i>August 2008 – July 2012</i>
---	--

Fuzhou University <i>Bachelor of Science in Applied Physics</i>	Fuzhou, China <i>September 2004 – June 2008</i>
---	--

Professional Skills

- **Programming Languages:** C/C++, Python, dotNet, F sharp, Bash, HTML, CSS, Javascript, SQL, Objective C, Matlab, VHDL, PLC, Assembly, LaTeX, LabView
- **OS/Middleware:** Linux, Windows, FreeRTOS, ROS
- **Software Framework/Libraries:** Qt, OpenGL, Protobuf, Boost, Matplotlib, Numpy, Scipy, Pcl, Eigen, Sophus, G2O, Ceres, OpenCV, NodeJs, Pytorch, Tensorflow
- **Software Tools:** CMake, SCon, Conan, Gitlab, Jenkins, Docker, AWS, Git, Confluence, Jira, Office365, QGIS
- **Technical Stack:** Ultrasonic/Radar/Video/Lidar knowledge, Embedded Software Design, Data Analysis and Visualization, modelling and simulation, SLAM, Sensor Fusion, Nonlinear Optimization, Machine learning, Deep learning, Computer Vision